

Activity Patterns and Speeds of Large Cruise Ships in Southeast Alaska

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We characterized activity patterns of large cruise ships in Southeast Alaska, where there is considerable overlap with aggregations of endangered humpback whales and where lethal ship–whale collisions have been reported. Using Automatic Identification System (AIS) data from 451 voyages representing 26 cruise ships ($\bar{x}$ length = 270 m), we summarized traffic intensity, speed, and timing (day vs. night) for the 2012 summer cruise season in Southeast Alaska. Traffic intensity was found to exceed 700 transits in some areas, which constituted up to 10 ships transiting these areas per day during peak season. Throughout most of the study area (68% of 5×5 km cells) average ship speed was >16 knots, and often >18 knots. About half (47%) of the time spent cruising occurred at night when detection and avoidance of whales by ships is likely low. Although whales in Southeast Alaska were subject to low levels of shipping traffic relative to other major shipping routes, cruise ships transited throughout the majority of our study area at speeds where the probability for lethality, should a collision occur, is high, potentially offsetting the relatively low traffic levels. Our results help identify areas of greatest relative risk to whales and locations where whale conservation measures may be most effective.

Keywords cruise ship traffic, humpback whale, ship speed, ship–whale collisions

Introduction

The world shipping fleet has changed dramatically in the past several decades with both the volume and average speed of ships increasing substantially (Ross 2005; McDonald, Hildebrand, and Wiggins 2006; Buhaug et al. 2009). Coinciding with these changes has been an emerging awareness of the deleterious impacts that shipping may have on individuals and populations of large whales (e.g., NRC 2005; Tyack 2008). Chronic, low-frequency underwater noise produced by large ships, which has been increasing for decades (Andrew, Howe, and Mercer 2002) has been attributed to loss of acoustic communication space (Hatch et al. 2012) and increased stress levels (Rolland et al. 2012) for species of baleen whales who use sound to communicate, navigate, and detect predators and prey (Clark et al. 2009). Large ships may also have direct fitness impacts to individuals or populations of whales via collisions (“ship strikes”; Laist et al. 2001). Collisions have been documented in virtually every ocean where both large ships and large whales overlap (Laist et al. 2001; Jensen and Silber 2003; Panigada et al. 2006; Neilson et al. 2012), and are frequent enough

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to inhibit population recovery for some species, such as the North Atlantic right whale (Fujiwara and Caswell 2001; Kraus et al. 2005).

Nevertheless, impacts of shipping on whale “life functions” or activities are difficult to measure, particularly for large cetaceans resulting in significant uncertainty. For example, while whales in or near shipping lanes are subject to chronic shipping noise resulting in communication “masking” (Hatch et al. 2008; Clark et al. 2009; Allen et al. 2012; Hatch et al. 2012), uncertainty in baleen whale hearing capacity, and an incomplete understanding with which whales encounter ships (Ellison et al. 2012), makes it difficult to link the sound exposure levels to fitness-related impacts (NRC 2005; Nowacek et al. 2007). Ship–whale collisions, in contrast, have a much more direct linkage to (individual) fitness, as many collisions with large ships are lethal (Vanderlaan and Taggart 2007; Conn and Silber 2013). However, estimating the rate of collisions and thus understanding population impacts, remains difficult owing to low detection probability (Kraus 2005) and because few shipping entities have mandatory requirements should a collision be detected.

Despite these uncertainties, characteristics of ships and shipping can provide insight into the risk of impacts to individuals and populations of whales and their habitat, as well as provide metrics by which conservation measures may be assessed (Wiley et al. 2011). Fundamental is quantifying the volume of shipping in important whale-use areas with the common sense assumption that increases in spatial/temporal overlap increases acoustic impacts or collision risk. For example, Automatic Identification System (AIS) data on ship volume and location has demonstrated that re-routing shipping traffic around high-use whale habitat reduces absolute or relative risk to whales by more than 80% in the Roseway Basin on the Scotian Shelf (Vanderlaan et al. 2009; van der Hoop, Vanderlaan, and Taggart 2012), up to 90% in the Gulf of Panama (Guzman et al. 2012), and over 40% for vessels nearing major ports along the southeastern coast of the United States (Fonnesbeck et al. 2008). Ship speed is another important risk factor, as the lethality of ship–whale collisions increases with increasing ship speed, with most collisions resulting in severe injury or death of whales when ships are travelling at ≥ 16 knots (Vanderlaan and Taggart 2007; Conn and Silber 2013). Faster ships may also increase the chance of a collision due to hydrodynamic draw should a whale surface very close to the ship (Silber, Slutsky, and Bettridge 2010) and may result in higher frequency of closer encounters by ships, either because whales have less time to detect and avoid ships or because ship operators have less time to detect whales and enact avoidance measures (Gende et al. 2011). Several recent studies have demonstrated significant decreases in the number of ship–whale collisions in areas where speed is regulated to protect right whales on the U.S. Atlantic coast (Conn and Silber 2013; Laist, Knowlton, and Pendleton 2014; van der Hoop et al. 2014). Thus, understanding ship speed provides not only an additional metric of risk but also a measure for assessing the efficacy of conservation measures should speed restrictions be enacted (Wiley et al. 2011).

Here we characterize shipping intensity and speed for large ($\bar{x}$ length = 270 m, range = 173–294 m) cruise ships in Southeast Alaska, an area that supports a significant fraction of the central North Pacific stock of humpback whales (*Megaptera novaeangliae*) (Calambokidis et al. 2008; Hendrix et al. 2012) currently listed as an endangered species under the U.S. Endangered Species Act. While large cruise ships represent a relatively small fraction of the global commercial shipping fleet (Buhaug et al. 2009), their annual growth has exceeded 7% since 1980 (CLIA 2012), and they are a predominant type of large ship in southeast Alaska (NRPG 2012). Although cargo and tanker ships are a predominant ship in other coastal regions (McKenna et al. 2012; Guzman et al. 2012; Wiley et al. 2011), there are no ports that accommodate these types of ships in southeast Alaska, and communities

in the area are primarily serviced by tank and dry cargo freight barges. Freight ship activity from log and ore carriers is also limited in southeast Alaska due to mill closures or limited capacity at ore terminals (NRPG 2012). And while there are large (> 100 m) ferries operating year round, there are more than four times the number of cruise ships than large ferries transiting the area during the summer cruise season (May to September) (NRPG 2012). Cruise ships have also been linked to a number of lethal collisions with whales both in this area and other areas of Alaska (Neilson et al. 2012).

In addition to intensity and speed, we also quantified whether transits occurred during daylight hours or at night. In the Northwest Atlantic and Mediterranean, there have been increased efforts to communicate a warning to mariners when whales are in an area with the stated goal to “... help ships avoid whales” (see: listenforwhales.org; REPCET.org). The implicit assumption is that mariners must be able to detect whales in order to effectively implement avoidance measures. Thus understanding the degree to which transits occur at night helps understand the likelihood that ship pilots/captains are able to visually detect whales during a transit.

Our objectives were to better understand the characteristics of the shipping traffic that humpback whales are exposed to in Southeast Alaska to help identify potential areas of greatest risk based on factors important to whale disturbance or potential collisions. Our analysis was focused on transit volume, routes (location), speed, and timing (day vs. night) of cruise ships for all of Southeast Alaska.

Methods

Study Area

Alaska is a popular cruise destination attracting nearly 1 million cruise passengers annually with the majority of port visits occurring in the southeastern portion of the state (i.e., “Southeast Alaska”). Southeast Alaska is an archipelago of islands, mainland fjords, and relatively narrow straits (Figure 1) with intense oceanographic mixing that result in high levels of primary and secondary productivity, productive forage fish communities, and abundant marine mammal populations (Etherington et al. 2007; Weingartner et al. 2009; Matthews et al. 2011; Hendrix et al. 2012; Womble and Gende 2013). The five-month summer cruise season of May to September corresponds to the summer feeding season for humpback whales. Whales generally arrive in Southeast Alaska in the spring (April/May) following migrations from wintering areas, primarily in Hawaii, and generally remain in the area until the fall (September–November) when they migrate back to Hawaii. Once in Southeast Alaska, whales are patchily distributed and movement among areas is temporally dynamic (Calambokidis et al. 2008; Hendrix et al. 2012). While no systematic surveys exist with regards to total number of whales in the area, several areas with consistent aggregations of whales have been subject to long-term photographic mark re-sight efforts, including within Frederick Sound, Glacier Bay National Park and adjacent Icy Strait, and Sitka Sound (Hendrix et al. 2012). These data have been used to generate a recent abundance estimate of 1585 whales with a growth rate of approximately 5% per year since 1986 (Hendrix et al. 2012). However, these data do not provide unbiased estimates of and thus variation of whale densities/use across the study area.

Our study area included waters up to 32 km from shore and was bounded to the south by the U.S. maritime boundary in Dixon entrance and to the north by Cape Spencer (Figure 1). We considered a distance of 32 km from shore to be biologically meaningful because

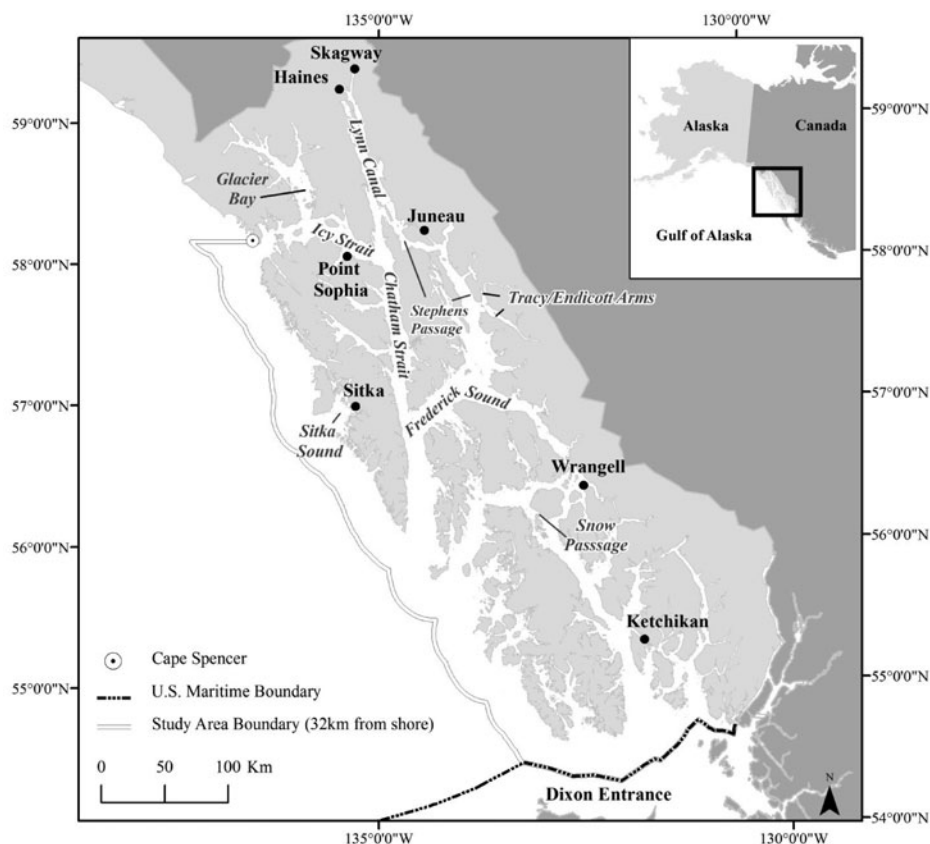


Figure 1. The study area included inside waters out to 32 km from shore and was bounded by the U.S. Maritime Boundary to the south and Cape Spencer to the north.

humpback whales aggregate in the near shore and inside waters of Southeast Alaska and we assumed full AIS coverage within this range (MXAK 2012).

AIS Data and Cruise Ships

All passenger vessels of 150 gross tonnage or greater transiting U.S. waters are required to carry AIS transponders (33 CFR §164.46—Federal Register 2003). AIS transponders send autonomous VHF signals including ship name, geographic location, speed, and course to shore-based or ship-based AIS receivers several times a minute (Tetreault 2005). AIS data from cruise ships was obtained from the Marine Exchange of Alaska. The Marine Exchange of Alaska is a non-profit maritime organization funded in part by the United States Coast Guard, state of Alaska and the marine industry, and was established to meet the needs of the Alaska maritime community by providing information, communications, and services that ensure safe, secure, efficient, and environmentally responsible maritime operations (MXAK 2012). AIS information was obtained using the Marine Exchange of Alaska's secure vessel tracking program, Pactracs 2.0. We used the published Cruise Line Agencies of Alaska cruising schedule to guide our search criteria in Pactracs 2.0, which ensured we included all AIS data from cruise ships in our analysis (CLAA 2012). AIS data

included ship location (latitude and longitude) in one minute intervals, time (UTC time and local time), Maritime Mobile Service Identity (MMSI) number, International Maritime Organization (IMO) number, call sign, ship name, speed over ground, and course over ground. Cruise ship dimensions were obtained from the cruise company (CCL 2013; CC 2013; DCL 2013; HAL 2013; NCL 2013; PC 2013; RCI 2013; RSSC 2013; SC 2013). We considered a “voyage” or “cruise” as a single trip by a ship within Southeast Alaska, which often lasted multiple days.

Ship Traffic Intensity

We used ship traffic intensity as our first measure of risk. Ship traffic intensity was quantified as the frequency of ship transits at speeds of 1 knot or faster through a 5×5 km grid cell overlaid on our study area. To ensure all possible grid cells for which cruise ships transited were included in our analysis, we used ArcGIS 10.1 Tracking Analyst (ESRI 2012) to convert AIS data points into lines. Separation distances that led to erroneous results (e.g., cruise ship paths intersecting land) were eliminated from our analysis (0.06% of the total data). In some cases, the ship’s path bisected a cell whereas in other instances the ship’s path may only cover a corner of a cell. While we counted these as equal in terms of traffic intensity in that cell, we made inferences across larger spatial scales (at the level of straits or passages, i.e., groups of many cells) rather than comparisons among individual cells, and thus believe we were able to capture the variation in traffic intensity across the study area in this manner.

Ship Speed

Mean ship speed over ground (SOG) was used to characterize speed patterns in our study area. We first calculated mean ship transit speed in a grid cell for a unique voyage and then used these values to calculate mean ship speed per grid cell for all voyages combined. We summarized ship speed in 2-knot increments, each of which have relative management thresholds including 10 knots (Seasonal Management Areas in the Northwest Atlantic; 50 CFR §224.105—Federal Register 2008), 12 knots (probability of severe encounters, probability of lethal collisions; Vanderlaan and Taggart 2007; Gende et al. 2011), 14 knots (Laist et al. 2001) and 16 knots (when effective probability of lethality reaches 1; Vanderlaan and Taggart 2007). Using mean speed per voyage per grid cell, we calculated mean ship speed in Southeast Alaska. We also examined the proportion of time ships spent cruising in each speed category. Data were evaluated for normality from histograms and quantile plots. When data did not meet assumptions of normality, we reported both mean and median values.

Transit Timing

Because the ability for ship operators to detect whales is limited or impossible at night, we also characterized ship traffic patterns by night. Although there are a number of factors that may limit the ability of ship operators to detect whales during the day (e.g., sea state and visibility), cruise ships often make port visits during the day and travel among ports at night. We thus examined ship traffic intensity when ships operated at night to identify areas where whales might be exposed to ships under operating conditions where captains and pilots are unable to detect whales and thus take whale avoidance measures. While speed is less important at night for detection, we nevertheless also separated out speeds at night

if the mechanism for reduction in collision is due to whales detecting and avoiding ships rather than pilots and captains detecting and avoiding whales. Although there is a period of time after sunset and before sunrise where there is enough light for ship operators to detect whales, the extent with which this may vary under clear sky versus overcast conditions is difficult to quantify, and for this reason, we defined night as the period of time between sunset and sunrise. Sunset and sunrise times varied throughout the season with 16 available daylight hours at the beginning of May, 18 by mid-June, and 12 by the end of September.

Results

A total of 26 ships were included in our analysis, representing all large main-line cruise ships (length range: 173–294 m) to visit Southeast Alaska in 2012 (Table 1). Ship visits constituted 456 voyages into the study area from the time period May 2 through September 27, 2012 (149 days). Voyages peaked during the month of July at 113 ($\bar{x} = 15.3$ ships in southeast Alaska/day), but traffic intensity was also high in June ($n = 104$; $\bar{x} = 14.8$ ships/day) and August ($n = 104$; $\bar{x} = 15.1$ ships/day), with fewer voyages in May ($n = 68$; $\bar{x} = 9.1$ ships/day) and September ($n = 67$; $\bar{x} = 11.1$ ships/day). The number of cruise ships present in our study area on any given day ranged from 2 to 18 in May to 9 to 21 in July. Most ships conducted at least 12 separate voyages to Southeast Alaska during 2012 (Table 1).

Cruise ships spent, on average, 94.1 hours (median = 97.0; ~ 4 days) in our study area per voyage. The majority of time ($\bar{x} = 65\%$) was spent cruising (moving ≥ 1 knot) with the remainder of time spent berthing (at port) or maneuvering (moving < 1 knot) (Table 2). While we had consistent AIS coverage for most of the study area, five voyages were excluded from our analysis due to incomplete data points across a significant portion of the cruise path. Thus our analysis of ship traffic intensity and ship speed comprised 451 voyages (98.9% of total voyages).

Ships generally followed the same routes connecting major ports of call such as Ketchikan, Juneau, and Skagway, and wilderness fjord destinations including Tracy and Endicott Arms and Glacier Bay National Park (Table 2, Figure 2A); there were few areas in Southeast Alaska that were not subject to cruise ship traffic (Figure 2A). Because ships generally followed the same route for each voyage, with some waterways relatively narrow and with some ports being visited more frequently than others (Table 2), there was high variation among 5×5 km cells. For example, Lynn Canal and portions of Stephens Passage, both waterways connecting major ports of call of Skagway and Juneau, were subject to more than 700 transits by cruise ships constituting up to 8 and 10 transits per day. In contrast to these high intensity traffic areas, the areas near Sitka were subject to 150 transits constituting up to 4 transits per day (Figure 2A).

Most areas within Southeast Alaska, regardless of volume, were transited by cruise ships at average speeds exceeding 16 knots (range = 16.1 to 22.2 knots) (Figure 2B). Forty-one percent of the cells had average speeds of 18 knots or more and 68% had average speeds of 16 knots or more, demonstrating that if whales are in or near a shipping route, they will be subject to lethal ship speeds should a collision occur. In contrast, 0.02% of cells had average speeds of 10 knots or less, most of which were near ports of call, in tidewater glacier viewing areas, or in lower Glacier Bay where ship speed is regulated and ships slow to embark and disembark NPS interpretive rangers. Of the time spent cruising, ship speed averaged 16.8 knots (± 2.7 SD; median = 17.4). Nearly 30% of all time spent cruising

Table 1

Ship names and dimensions for main line cruise ships transiting Southeast Alaska during cruise season 2 May through 27 September, 2012. Voyages are the total number of cruises per ship during the cruise season (CLAA 2012). Three cruise ships were excluded from analysis because they comprised <1% of the total voyages made and their specialized cruise itineraries were a departure from those offered by the main line cruise companies.

Ship	Company	Length (m)	Tons	Voyages
Amsterdam	Holland America Line	238	62,735	18
Carnival Spirit	Carnival Cruise Lines	293	88,500	20
Celebrity Century	Celebrity Cruises	248	71,545	19
Celebrity Infinity	Celebrity Cruises	294	91,000	19
Celebrity Millennium	Celebrity Cruises	294	91,000	15
Coral Princess	Princess Cruises	294	91,627	18
Dawn Princess	Princess Cruises	261	77,441	1
Diamond Princess	Princess Cruises	290	115,875	19
Disney Wonder	Disney Cruise Line	294	83,000	16
Golden Princess	Princess Cruises	290	108,865	18
Island Princess	Princess Cruises	294	92,000	19
Norwegian Jewel	Norwegian Cruise Line	294	93,502	18
Norwegian Pearl	Norwegian Cruise Line	294	93,530	20
Oosterdam	Holland America Line	285	82,305	21
Radiance of the Seas	Royal Caribbean International	293	90,090	19
Rhapsody of the Seas	Royal Caribbean International	279	78,491	18
Sapphire Princess	Princess Cruises	290	115,875	19
Sea Princess	Princess Cruises	261	77,499	12
Seven Seas Navigator	Regent Seven Seas Cruises	173	28,803	14
Silver Shadow	Silversea Cruises	186	28,258	14
Star Princess	Princess Cruises	290	108,977	19
Statendam	Holland America Line	219	55,819	18
Volendam	Holland America Line	238	61,214	21
Westerdam	Holland America Line	285	82,348	20
Zaandam	Holland America Line	238	61,396	21
Zuiderdam	Holland America Line	285	82,305	20
Total				456

occurred at >18 knots and nearly 50% of the time ships cruised at ≥ 16 knots (Figure 3). In contrast, ships spent 10% of the time cruising at speeds of 10 knots or less (Figure 3).

Transit Timing

When in Southeast Alaska, about half of the cruising occurred at night (47%) although the spatial distribution of ship traffic intensity and speed shifted depending on transit timing reflecting differences in the target port of call (Figures 4 and 5). Much of the night-time traffic occurred in Lynn Canal as ships transited from Juneau to Skagway and from Skagway to other destinations such as Icy Strait en route to Glacier Bay. However, much of the night-time traffic speed in these areas was slow, averaging around 12 knots (Figure 5).

Table 2

Number of port visits (*n*) and proportion of time spent in hours (% time) on a per voyage basis in port or at a wilderness fjord, while cruising or berthing, and for all voyages during the study period 2 May 2–27 September 27, 2012. The % time for ships in port or wilderness fjord is relative to the total time spent at each location. The % time for ships underway versus in port and for all voyages during the study period are relative to the total time spent in the study period. (Note: TEA – Tracy/Endicott Arms).

		<i>n</i>	% time (h)
Port	Haines	21	1.7%
	Juneau	442	27.0%
	Ketchikan	427	19.7%
	Point Sophia	61	3.0%
	Sitka	77	3.9%
	Skagway	348	26.9%
	Wrangell	1	0.1%
Wilderness Fjord	Glacier Bay	207	12.4%
	TEA	180	5.4%
Operating Status	Cruising	451	65.0%
	Berthing	451	35.0%
All voyages	—	451	

Discussion

We analyzed large cruise ship traffic patterns to help understand the levels of shipping exposure to the population of endangered humpback whales in Southeast Alaska. Although our study comprised only one year of data, between 25 to 28 large cruise ships (>150 m) have made more than 400 voyages to Southeast Alaska every summer since 2010 (CLAA

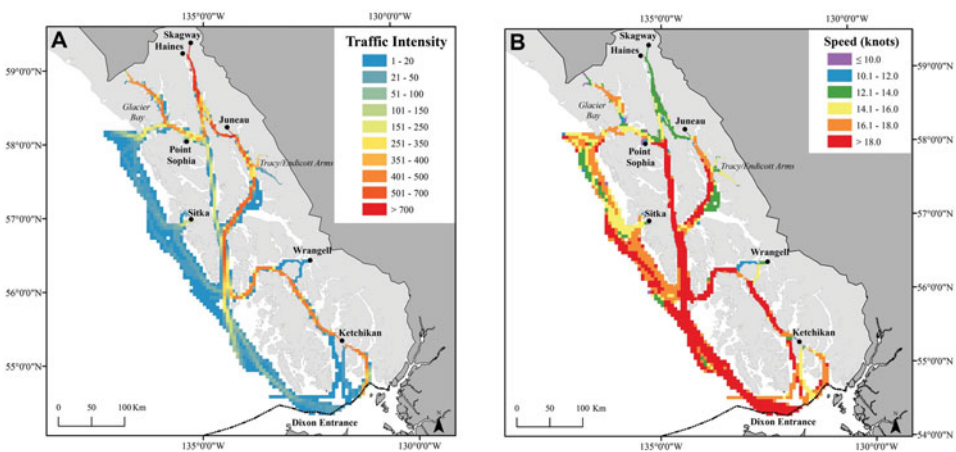


Figure 2. Cruise ship traffic intensity (A) and speed (B). Traffic intensity is represented as the total number of unique voyage transits per grid cell. Speed is represented as the mean speed over ground for all unique voyages per grid cell.

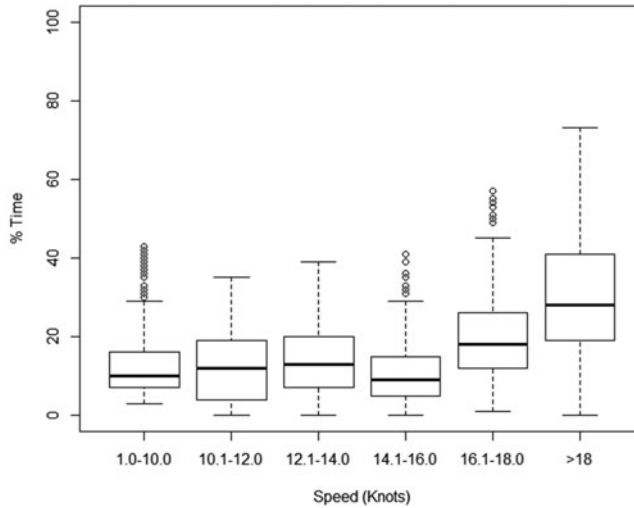


Figure 3. Boxplot of the % time large cruise ships spent cruising in each speed category in Southeast Alaska, May 2—September 27, 2012. The box represents the range in the proportion of time spent in each speed category between the 25th and 75th percentiles, the black center bar represents the median, and the range in whiskers represent 1.5 times the interquartile range above and below the 25th and 75th percentiles. Outliers are represented as points that fell beyond the whisker range.

2010; 2011), with similar numbers in Southeast Alaska during the 2013 cruise season (CLAA 2013; JCVB 2013). Thus, the number of cruise ships and voyages reported during 2012 reflects the volume of shipping traffic in this area in recent summers and likely reflects future shipping conditions.

Our results identified significant spatial variation in cruise ship traffic throughout Southeast Alaska with a primary, high-intensity route connecting the ports of Ketchikan,

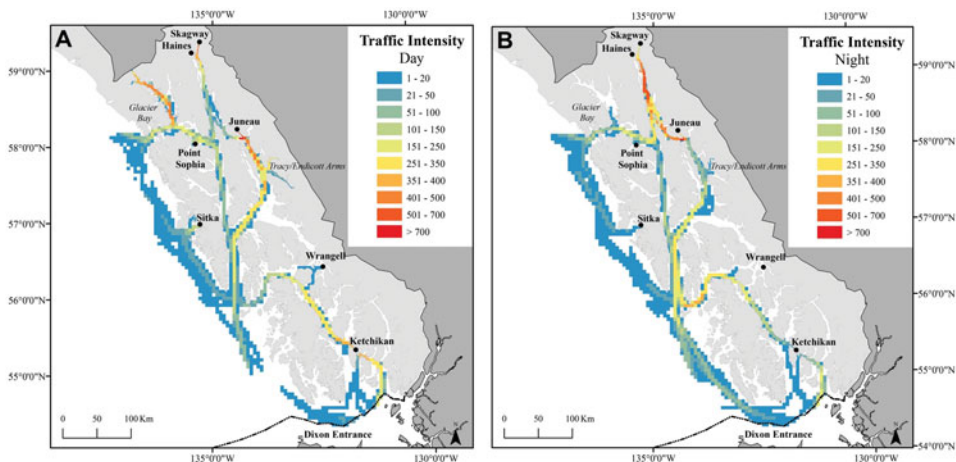


Figure 4. Traffic intensity by day (A) and night (B). Traffic intensity is represented as the total unique voyage transits per grid cell. Day and night are based on sunset and sunrise times, which vary throughout the season.

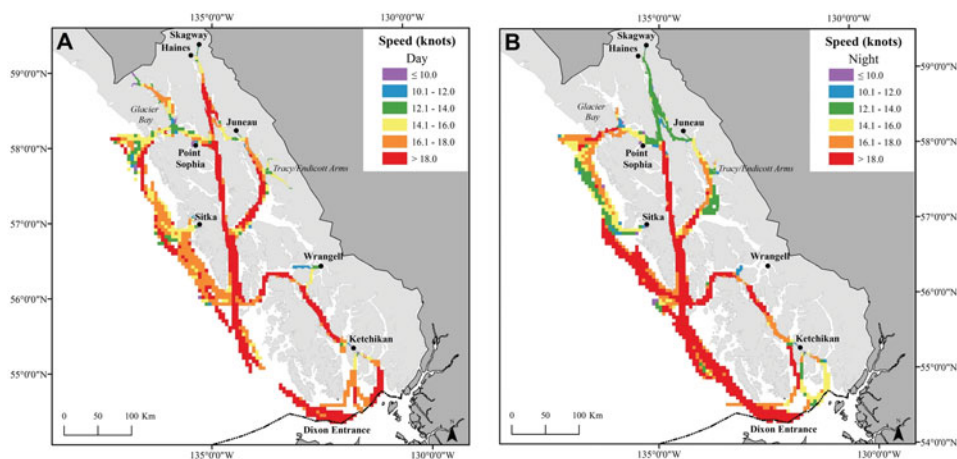


Figure 5. Ship speed by day (A) and night (B). Speed is represented as the mean speed over ground for all unique voyages. Day and night are based on sunset and sunrise times, which vary throughout the season.

Juneau and Skagway, and to a lesser extent the wilderness fjords of Tracy/Endicott Arm and Glacier Bay. This route, and particularly Frederick Sound, Stephens Passage, and Lynn Canal, had transit intensity exceeding 500 in some 5×5 km cells, with as many as seven to ten ships transiting these areas per day during peak season. Nearly 500 individually identified humpback whales were documented using Frederick Sound as of 2008, with perhaps up to 700 whales as of 2012 assuming a median rate of 50 new whales identified per year based on photographic mark/re-sight surveys (Hendrix et al. 2012). Thus, while Frederick Sound has long been known to be a whale aggregation hotspot, our results identify it as an area where collision risk may be particularly high based on ship transit intensity. The lack of systematic, area-wide whale surveys nevertheless makes it impossible to infer differences in relative risk among these areas.

Much of the high-intensity route between Ketchikan and Endicott/Tracy Arm was also characterized by average ship speeds exceeding 18 knots. Speed is a prominent risk factor for lethal ship–whale collisions with faster ships having (a) a higher probability of closer encounters, with encounter distances on average over 100 m closer for ships travelling faster than 11.8 knots (Gende et al. 2011); (b) a higher probability of collision should whales surface very close to the ship, owing to hydrodynamic forces (Silber, Slutsky and Bettridge 2010); and (c) an increased probability that a collision will be lethal should one occur (Vanderlaan and Taggart 2007; Conn and Silber 2013). At 18 knots or more, nearly 100% of collisions are expected to be lethal (Vanderlaan and Taggart 2007). While we cannot infer whether lethal collision risk is higher for areas that experience high transit/low average speeds, such as Stephens Passage and Lynn Canal (average speeds ~ 14 knots), compared to the low intensity/high speed areas, such as northern Chatham Strait (average speeds ~ 18 knots) and off the coast of Sitka (average speeds ~ 16 knots), we nevertheless conclude that the area between southern Chatham Strait and Tracy/Endicott Arms, which experiences both high relative traffic intensity and high average ship speeds, is an area of greatest risk for lethal collisions based on shipping characteristics.

While our results identified areas of high risk relative to other areas within our study area, it is difficult to infer whether whales in high traffic areas in Southeast Alaska, based

on shipping characteristics, are more or less at risk of lethal collisions compared to other areas of the United States where whale conservation measures have been proposed or implemented. The transit volume by area of cruise ships in Frederick Sound was much lower than in the Whale Advisory Zone in the Santa Barbara Channel (McKenna et al. 2012), but similar to transit volume by area in Stellwagen Bank National Marine Sanctuary (Wiley et al. 2011) and the Gulf of Panama (Guzman et al. 2012) (Table 3). Nevertheless, average speeds for cruise ships in Frederick Sound (16.4 knots) were faster than average speeds for large ships (mostly container and cargo vessels) transiting through the Stellwagen Bank National Marine Sanctuary (~average 13.5 knots; Wiley et al. 2011), the Roseway Basin (~14.6 knots; van der Hoop, Vanderlaan and Taggart 2012), and in Vessel Management Areas along the western North Atlantic where there was a relatively low probability of vessel speeds at or above 16 knots for cargo vessels (Figure 2 in Conn and Silber 2013). Given that the odds of a lethal injury to a whale, should a collision occur, increases 1.5 fold with every 1-knot increase in speed (Vanderlaan and Taggart 2007), the lower risk of lethal collisions due to fewer transits from cruise ships in areas like Frederick Sound may be offset somewhat by higher average speeds.

We are not aware of any studies that have tested the implications of day- versus night-time travel relative to the risk of lethal collisions. However, the risk of collisions will be elevated under any conditions that limit the ability of ship operators to detect whales, assuming ships can play an active role in whale avoidance. This assumption is supported by evidence that some species of whales appear to ignore ships entirely (Nowacek, Johnson and Tyack 2004), perhaps owing to an inability to pinpoint the source of ship noise when at or near the surface due to deviances in noise refraction near the surface, the presence of other traffic in the area, or bow null-effect acoustic shadow zones (Terhune and Verboom 1999; Laist et al. 2001; Allen et al. 2012). Based on hundreds of observations of humpback whales surfacing around cruise ships in Southeast Alaska we have demonstrated that whale responses are highly variable with some whales exhibiting last second flight responses while others appear unaware of the ship entirely, despite encountering the ship at very close distances (<75 m; Harris et al. 2012). Based on discussions with a number of pilots and captains on the bridge of cruise ships, the effectiveness of whale avoidance is contingent, in part, upon (1) their ability to detect whales at a distance sufficient to invoke an avoidance measure owing to limited maneuverability and (2) their ability to infer a whale's direction of travel to avoid course adjustments that could elevate collision risk. Each of these conditions is likely inhibited with limited visibility. The relatively high intensity transit areas in Lynn Canal and Stephens Passage, although transited at lower average speeds, mostly occurred at night. Thus, while we cannot quantify the extent to which risk is reduced for daytime travel versus night, particularly because factors such as sea surface conditions and visibility also likely influence detection probability during the day, we assume that traveling at night limits the visual ability of ships to detect whales and thus increases risk of collision.

Given that relative risk may be higher in certain areas within Southeast Alaska, a fundamental step is to address whether management actions are necessary, and if so, what management actions are feasible. To date, management options implemented to reduce collision risk between large ships and whales have focused on two alternatives: reducing spatio-temporal overlap by re-routing ships around high-use whale areas or by reducing ship speed in important whale habitat. While shifting shipping lanes or designating Areas To Be Avoided (ATBA) has proved an effective management strategy for reducing absolute or relative risk of lethal collisions in other areas (Fonnesbeck et al. 2008; Vanderlaan and Taggart 2009; Guzman et al. 2012; van der Hoop, Vanderlaan and Taggart 2012; Redfern et al. 2013), the feasibility for such a measure in Southeast Alaska is limited. Southeast

Table 3
Cruise ship transit volume in Frederick Sound, Alaska relative to other areas where whale conservation measures have been proposed or implemented.

Location	Vessel type	No. transits	Traffic Area area (km ²)	Traffic days	Transits per day	Transits per day/km ²
Whale Advisory Zone, Santa Barbara Channel, CA (Table 1 in McKenna et al. 2012)	Cargo and tanker	10,132	678	830	12.2	0.0180
Stellwagen Banks National Marine Sanctuary, MA (Table 2 in Wiley et al. 2011)	All	2,518	2,181	365	6.9	0.0032
Gulf of Panama (Table 1 in Guzman et al. 2012)	All	892	11,669	30	29.7	0.0025
Frederick Sound, AK	Cruise ships	524	1,300	149	3.5	0.0027

Alaska is characterized by narrow fjords, straits, and channels often less than 10 km across with some areas of known high whale concentrations, such as the mouth of Glacier Bay National Park, 5 km or less in width with an even narrower safe navigational channel. Thus, the ability to shift a shipping lane within a primary transit route is limited. Designating an alternative route to access the few ports of call within Southeast Alaska would also require surveys of whales in each of the alternatives, as even small aggregations of whales in narrow passageways can result in high densities and resultant elevated collision risk.

In sharp contrast, limiting ship speed may be a feasible management option. Currently there are no speed regulations in Southeast Alaska except in Glacier Bay National Park, where vessels are subject to speed restrictions of 20 knots in the lower bay throughout the season, or 13 knots in the lower bay or other areas of the park where whale aggregations persist long enough to warrant additional protective measures (36 CFR §13—Federal Register 2006). Throughout the rest of Southeast Alaska, the only requirement is for all mariners to operate at “safe, slow speeds” when near humpback whales (50 CFR§224.103—Federal Register 2001). Some voluntary speed restrictions to lower whale collision risk have been endorsed in our study area by the Southeast Alaska Pilots Association who has identified Snow Passage as an area where reduced speeds may be required due to channel restrictions and the presence of other traffic and whales in the area (NRPG 2012).

Yet, the willingness of ships to voluntarily reduce speed may be low when it may significantly impact revenue. Voluntary speed reduction measures have had minimal compliance for cargo and tanker ships in southern California (McKenna et al. 2012) and has shown to be strongly influenced by the threat of enforcement or citations along the Northwest Atlantic (Silber, Adams, and Fonnesbeck 2014). While fuel consumption constitutes a significant operational cost for large ships and is proportional to the third power of a ship’s speed (Ronen 2011), the “optimal” speed at which cruise ships travel is a complex function of distance to next port of call, emissions, engine loads, navigational hazards, and other factors, all of which must be considered relative to revenue opportunities. For example, following a port of call to Glacier Bay National Park, cruise ships commonly travel to Ketchikan, a distance of 278 nautical miles (514.9 km) and most often at speeds of 18 knots or more. While traveling this speed likely increases fuel consumption and results in higher risk to whales, the elevated speeds allow the ship to make port by morning enabling the ship to sell revenue-generating, shore-based excursions such as whale watching, flightseeing, charter fishing, and other tours. A speed regulation of 10 knots along this entire route would result in the ship arriving at 8 PM, long after they would be able to allow passengers to embark on revenue generating tours. Speed restrictions would thus likely be met with significant resistance both from cruise ship companies and the communities at the receiving ports. Alternatively, ships commonly travel between Juneau and Skagway, which is separated by only 100 nautical miles (185.2 km) and ships departing Juneau in the late evening only need to make 10 knots to arrive in Skagway early the next morning. Thus, the willingness of ships to voluntarily comply to speed restrictions may vary depending on the distance ships must travel to reach the next port of call.

Ultimately, the decision to generate new regulations should carefully consider the benefit for meeting whale conservation objectives with the potential economic costs to the industry and their respective ports of call. The cruise industry is an important economic component of the Southeast Alaska economy, generating significant in-port expenditures as part of bringing nearly 1 million visitors to Alaska annually (MG 2012). In the Northwest Atlantic, speed restrictions for large ships have been justified, in part, because the economic impacts to industry were offset by the conservation value of implementing speed restrictions, as the population abundance of North Atlantic right whales (~400 individuals) is critically

low and evidence demonstrates that even a few added mortalities from ship collisions can change the trend from stable to decreasing (Fujiwara and Caswell 2001). In Southeast Alaska the population abundance of humpback whales is comparatively high (~2,000 individuals) and has been growing by an average of 5% per year since 1986 (Hendrix et al. 2012), a period that also coincides with growth in the number of cruise ships visiting the area. While collisions between humpback whales and cruise ships have been documented on a number of occasions in the past (Neilson et al. 2012), the population growth rate may be able to “sustain” an increased number of collisions. The values-based decision should thus consider the economic impacts of speed reductions with its corresponding conservation impact to humpback whales in Alaska.

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